

$$(2) \quad 1 - \frac{1}{3^p} + \frac{1}{5^p} + \cdots + (-1)^{n+1} \frac{1}{(2n-1)^p} + \cdots;$$

$$(3) \quad \frac{1}{2\ln 2} - \frac{1}{3\ln 3} + \frac{1}{4\ln 4} + \cdots + (-1)^{n-1} \frac{1}{(n+1)\ln(n+1)} + \cdots;$$

$$(4) \quad \sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}-1}{n};$$

$$(5) \quad \frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \cdots + (-1)^{n-1} \frac{2n+1}{n(n+1)} + \cdots;$$

$$(6) \quad \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n!}{3^{n^2}};$$

$$(7) \quad \frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{3} \sin \frac{\pi}{3} + \frac{1}{4} \sin \frac{\pi}{4} - \cdots;$$

$$(8) \quad \sum_{n=1}^{\infty} (-1)^{n+1} \tan \frac{\varphi}{n} \quad \left(-\frac{\pi}{2} < \varphi < \frac{\pi}{2} \right);$$

(3) $\lim_{n \rightarrow \infty} u_n = 0$, $f(n) = (n+1) \ln(n+1)$, $f'(n) = \ln(n+1) + 1 > 0$, 故 $\{u_n\}$ 单调

收敛.
 $\sum_{n=1}^{\infty} \frac{1}{(n+1) \ln(n+1)}$ $f(n) = \frac{1}{(n+1) \ln(n+1)}$, $\lim_{A \rightarrow \infty} \int_0^A \frac{dn}{(n+1) \ln(n+1)} = \lim_{A \rightarrow \infty} \ln \ln(A+1)$, 无界

故条件收敛

(4)

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}-1}{n} = 0, \quad f(n) = \frac{\sqrt{n}-1}{n}, \quad f'(n) = \frac{\frac{n}{2\sqrt{n}} - \sqrt{n} + 1}{n^2} = \frac{1 - \frac{\sqrt{n}}{2}}{n^2}, \text{ 故 } \frac{\sqrt{n}-1}{n} \text{ 先增后减}$$

由于去掉有限项不影响敛散性, 故收敛

$$\frac{\sqrt{n}-1}{n} > \frac{1}{\sqrt{n}} - 1, \text{ 发散, 故条件收敛}$$

(6) $u_n = \frac{n!}{3^{n^2}}$, $\frac{u_n}{u_{n+1}} = n \cdot 3^{(n+1)^2 - n^2} = n \cdot 3^{1+2n} = \frac{n}{3^{2n-1}} < 1$, 故 u_n 单调, 且 $\lim_{n \rightarrow \infty} u_n = 0$, 收敛

$$\sum_{n=1}^{\infty} \frac{n!}{3^{n^2}}, \quad \lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = 0, \quad \text{故绝对收敛}$$

$$(10) \quad \sin(2\sqrt{n^2+1}) = \sin[(2(\sqrt{n^2+1}+n)-n)2] = \sin[2(\sqrt{n^2+1}+n)] \cos n2 - \cos[2(\sqrt{n^2+1}+n)] \sin n2 \\ = (-1)^n \sin[2(\sqrt{n^2+1}+n)]$$

$$u_n = \sin 2(\sqrt{n^2+1}+n) = \sin 2(\sqrt{n^2+1}-n) = \sin \frac{2}{\sqrt{n^2+1}-n}, \text{ 显然 } \frac{2}{\sqrt{n^2+1}-n} \text{ 在同一单调区间内}$$

故有 u_n 单减, 且 $\lim_{n \rightarrow \infty} u_n = 0$, 故收敛

2. 已知级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 证明级数 $\sum_{n=1}^{\infty} \frac{u_n}{n^p} (p > 0)$ 与 $\sum_{n=1}^{\infty} \frac{n}{n+1} u_n$ 均收敛.

3. 证明级数 $\sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^p} (0 < \varphi < 2\pi)$ 当 $p > 1$ 时绝对收敛, 当 $0 < p \leq 1$ 时条件收敛.

4. 研究级数 $\sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^p} \left(1 + \frac{1}{n}\right)^n (0 < \varphi < 2\pi, p > 0)$ 的敛散性.

5. 形如 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 的级数称作狄利克雷级数. 证明它有下列性质: 若级数 $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 收敛

(发散), 那么当 $x > x_0 (x < x_0)$ 时, 级数 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 也收敛(发散).

6. 设级数 $\sum_{n=1}^{\infty} u_n$ 绝对收敛, 证明级数 $\sum_{n=1}^{\infty} \frac{2n-1}{n} u_n$ 也绝对收敛.

2. $\sum_{n=1}^{\infty} u_n$ 收敛, 则部分和序列有界

而 $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ 由狄利克雷判别法, 级数收敛
且单减

$\frac{n}{n+1}$ 单减且 $\rightarrow 1$, 而 $\sum_{n=1}^{\infty} u_n$ 收敛, 由阿贝尔判别法, 级数收敛

3. $p > 1$ 时, $\sum_{n=1}^{\infty} \left| \frac{\cos n\varphi}{n^p} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^p}$, 绝对收敛

$0 < p \leq 1$ 时,

$$\therefore \left| \frac{\cos n\varphi}{n^p} \right| \geq \frac{\cos^2 n\varphi}{n^p} = \frac{1 + \cos 2n\varphi}{2n^p}, \quad \frac{1}{2n^p} \text{ 发散}, \quad \frac{\cos 2n\varphi}{2n^p} \text{ 收敛},$$

故绝对值后发散

而 $\frac{\cos n\varphi}{n^p}$ 中, $\frac{1}{n^p} \downarrow$ 且 $\rightarrow 0$. 故原级数收敛

$$5. \quad \frac{a_n}{n^x} = \frac{a_n}{n^{x_0}} \cdot \frac{1}{n^{x-x_0}}$$

若 $x > x_0$ 时, $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 收敛, 而 $x - x_0 > 0$, $\frac{1}{n^{x-x_0}}$ 单减, 且 $\frac{1}{n^{x-x_0}} \leq 1$ (有界)

由阿贝尔判别法, $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 收敛

而 $x < x_0$ 时, $\therefore \sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 发散, $\frac{a_n}{n^x} = \frac{a_n}{n^{x_0}} \cdot \frac{1}{n^{x-x_0}}$, $x - x_0 < 0$, 故 $\frac{1}{n^{x-x_0}} \geq 1$, $\frac{a_n}{n^x} > \frac{a_n}{n^{x_0}}$, 故原级数发散

6 $|\frac{2n+1}{n}| = 2 - \frac{1}{n}$ 单增且有 $\lim_{n \rightarrow \infty} \frac{2n+1}{n} = 2$.

故 $\sum_{n=1}^{\infty} \frac{2n+1}{n} |u_n|$ 收敛 (阿贝尔判别法)

故绝对收敛

$$\frac{1-x}{1+x} < 0$$

$$(1-x)(1+x) < 0$$



$$1 - \frac{2x}{1+x} > \frac{2}{1+x}$$

$$1-x < 1+x \quad 2x > 0$$

1. 求下列级数的收敛域:

(1) $\sum_{n=1}^{\infty} (\ln x)^n$; (2) $\sum_{n=1}^{\infty} \frac{1}{2n-1} \left(\frac{1-x}{1+x} \right)^n$; (3) $\sum_{n=1}^{\infty} \frac{1}{x^n} \sin \frac{\pi}{3^n}$.

1. (1) $0 < x < 1$ 时, $\frac{1-x}{1+x} > 0$, 故 $\sqrt[n]{u_n} = \left(\frac{1-x}{1+x} \right) \cdot \frac{1}{\sqrt[n]{2n-1}}$, $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \frac{1-x}{1+x}$, 若 $\frac{1-x}{1+x} < 1$, 即 $0 < x < 1$ 时收敛

而 $\frac{1-x}{1+x} = 1$, 即 $x=0$ 时, 易知发散

② $x=1$ 时, 易知收敛

③ $x > 1$ 时, $\left(\frac{1-x}{1+x} \right) < 0$, 为交错级数. $|u_n| = \frac{1}{2n-1} \cdot \left(\frac{x-1}{x+1} \right)^n$, $\sqrt[n]{u_n} = \frac{x-1}{x+1} \cdot \frac{1}{\sqrt[n]{2n-1}} \rightarrow \frac{x-1}{x+1} = 1 - \frac{2}{x+1} < 1$

故原级数收敛

④ $x < -1$ 时, $\frac{1-x}{1+x} < 0$, $|u_n| = \left(\frac{1-x}{-x-1} \right)^n \frac{1}{2n-1}$, $\sqrt[n]{u_n} = \frac{1-x}{-x-1} \cdot \frac{1}{\sqrt[n]{2n-1}} \rightarrow \frac{x-1}{x+1} = 1 - \frac{2}{x+1} > 1$

故收敛域为 $(0, +\infty)$